7. If the quotient obtained on dividing $3x^3 + 11x^2 + 34x + 106$ by $x - 3$ is $3x^2 + ax + b$, then find a, b and also the remainder.

3.6.3 Factorisation using Synthetic Division

In this section, we use the synthetic division method that helps to factorise a cubic polynomial into linear factors. If we identify one linear factor of cubic polynomial $p(x)$ then using synthetic division we can get the quadratic factor of $p(x)$. Further if possible one can factorise the quadratic factor into linear factors.

Note

- For any non constant polynomial $p(x)$, $x = a$ is zero if and only if $p(a) = 0$
- $x-a$ is a factor for $p(x)$ if and only if $p(a) = 0$ (Factor theorem)

To identify $(x-1)$ and $(x+1)$ are the factors of a polynomial

- $(x-1)$ is a factor of $p(x)$ if and only if the sum of coefficients of $p(x)$ is 0.
- $(x+1)$ is a factor of $p(x)$ if and only if the sum of the coefficients of even power of x, including constant is equal to the sum of the coefficients of odd powers of x

Example 3.38

(i) Prove that $(x-1)$ is a factor of $x^3 - 7x^2 + 13x - 7$

(ii) Prove that $(x+1)$ is a factor of $x^3 + 7x^2 + 13x + 7$

Solution

(i) Let $p(x) = x^3 - 7x^2 + 13x - 7$

Sum of coefficients $= 1 - 7 + 13 - 7 = 0$

Thus $(x-1)$ is a factor of $p(x)$

(ii) Let $q(x) = x^3 + 7x^2 + 13x + 7$

Sum of coefficients of even powers of x and constant term = 7 + 7 = 14

Sum of coefficients of odd powers of x = 1 + 13 = 14

Hence, $(x+1)$ is a factor of $q(x)$

Example 3.39

Factorise $x^3 + 13x^2 + 32x + 20$ into linear factors.

Solution

Let, $p(x) = x^3 + 13x^2 + 32x + 20$

Sum of all the coefficients $= 1 + 13 + 32 + 20 = 66 \neq 0$

Hence, $(x-1)$ is not a factor.

Sum of coefficients of even powers and constant term = 13 + 20 = 33

Sum of coefficients of odd powers = 1 + 32 = 33

Hence, $(x+1)$ is a factor of $p(x)$

Now we use synthetic division to find the other factors

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